Baipilli Naveen

Climate Risk — Geospatial AI — Earth Observation — Machine Learning

baipallinaveen60@gmail.com | +91-7995860489 |
linkedin.com/in/baipilli-naveen |
github.com/bapallinaveen60
bapallinaveen60.github.io/naveen-portfolio/

PROFESSIONAL SUMMARY

Climate and geospatial AI researcher/engineer with an M.Tech in Climate Change from IIT Hyderabad and a B.Tech in Geo-Informatics, working at the intersection of **Earth observation, environmental modelling, spatial data analysis, and machine learning**. Experienced with **INSAT-3DR, GPM DPR, ERA5, SRTM, Sentinel-1/2, and large-scale environmental datasets**. M.Tech research focused on physics-aware deep learning for precipitation by combining satellite, atmospheric, and terrain information, including evaluation under unseen temporal conditions. Experienced in **Python, PyTorch, TensorFlow, U-Net, Random Forest, XGBoost, GIS, spatiotemporal analysis, statistical methods, and large-scale environmental data processing**. Interested in developing **hybrid AI methods for climate extremes, flood and hydrological risk, spatial risk modelling, and environmental decision support**.

EDUCATION

Indian Institute of Technology Hyderabad – **M.Tech in Climate Change** 2024 – 2026
CGPA: **9.2/10**

Andhra University – **B.Tech in Geo-Informatics** 2019 – 2023
CGPA: **7.83/10**

RESEARCH INTERESTS

Climate Risk and Extreme Events | Flood and Hydrological Risk | Geospatial AI | Earth Observation | Hybrid Physics–AI Modelling | Spatiotemporal Machine Learning | Climate Change | Spatial Risk Modelling

RESEARCH EXPERIENCE

Physics-Aware Hierarchical Deep Learning for Multi-Class Precipitation Classification Using INSAT-3DR Brightness Temperature, Atmospheric, and Terrain Data M.Tech Thesis – IIT Hyderabad
Live Project: PrecipCast

- Developed a multimodal Earth-observation framework integrating **INSAT-3DR, GPM DPR, ERA5, and SRTM** to investigate and classify precipitation systems over India.
- Built end-to-end geospatial pipelines for satellite geolocation, calibration, spatio-temporal collocation, atmospheric-data matching, spatial tiling, quality control, and model-ready dataset generation.
- Developed a hierarchical **U-Net** framework for rain detection, precipitation-type discrimination, and fine-grained precipitation classification under strong class imbalance.
- Integrated **brightness temperature, elevation, slope, CAPE, and TCWV** to examine how atmospheric state and terrain provide physical context for data-driven precipitation classification.
- Evaluated temporal generalization using 2019–2023 for training and the independent 2024 monsoon season for testing; physics-aware inputs improved precipitation-type classification accuracy from **68.5% to 72.85%**.
- Investigated model behaviour using **confidence analysis, spatial error analysis, permutation-based feature importance, Layer Normalization, and attention-gated skip connections**.

PROFESSIONAL EXPERIENCE

Member of Technical Staff – Hornbill Ag Current

- Developing **PondSense**, a satellite-driven environmental intelligence platform integrating **Sentinel-1, Sentinel-2, ERA5, GIS features, and machine learning** for pond-scale environmental prediction and risk assessment.
- Building geospatial feature-engineering workflows combining **optical, SAR, meteorological, spatial, and ground-observation data** for environmental modelling.
- Developing computer-vision pipelines for agricultural applications using **object detection, image classification, and deep learning**.
- Working across geospatial preprocessing, model development, validation, inference, API integration, and deployment using **Python, Docker, Linux, FastAPI, Git, and cloud infrastructure**.

Geospatial Engineer – LTIMindtree, Chennai Sep 2023 – Jun 2024

- Processed **high-resolution drone remote-sensing imagery** to generate orthomosaics, DEMs, and derived geospatial products.
- Performed **terrain analysis, elevation modelling, raster/vector preprocessing, feature extraction, and geospatial data validation** using Python and GIS workflows.
- Developed repeatable geospatial processing workflows supporting mapping, environmental analysis, and downstream machine-learning applications.
- Worked with large spatial datasets and automated preprocessing and quality-control steps for geospatial analysis.

SELECTED CLIMATE & SPATIAL RISK PROJECTS

Dengue Hotspot Prediction Using Spatiotemporal Machine Learning IIT Hyderabad

- Developed **Random Forest and XGBoost** models using spatial, demographic, environmental, and climatic variables for hotspot prediction.
- Performed geospatial feature engineering using **NDVI, LST, water indices, and terrain variables** and generated **spatial risk maps**.

Regional Arctic Sea Ice Variability and Teleconnections with Climate Indices IIT Hyderabad

- Analysed four decades of satellite-derived Arctic sea-ice data using **wavelet and time-series analysis** to characterize regional climate variability.
- Investigated non-stationary relationships between sea ice and **ENSO, AO, and NAO** climate indices using cross-wavelet and wavelet-coherence methods.

TECHNICAL SKILLS

Programming	:	Python, NumPy, Pandas, SQL
Machine Learning	:	Scikit-learn, Random Forest, XGBoost, Feature Engineering, Model Evaluation
Deep Learning	:	PyTorch, TensorFlow, CNN, U-Net, Attention Mechanisms, Computer Vision, Image Segmentation
Earth Observation	:	INSAT-3DR, GPM DPR, Sentinel-1/2, ERA5, SRTM, Drone Remote Sensing
Geospatial	:	GeoPandas, Rasterio, GDAL, QGIS, ArcGIS Pro, PostGIS, Raster/Vector Analysis
Scientific Data	:	xarray, Zarr, NetCDF, HDF5, GeoTIFF, Spatiotemporal Data Processing
Research Methods	:	Statistical Analysis, Time-Series Analysis, Wavelet Analysis, Spatial Error Analysis, Feature Importance, Model Interpretation, Generalization Evaluation
Software / MLOps	:	Git, Docker, Linux, FastAPI, Cloud Deployment

PUBLICATIONS & RESEARCH OUTPUTS

Peer-Reviewed Conference Publications

1. M. Saranya, **B. Naveen**, and R. Maheswaran, "Regional Arctic Sea Ice Variability and Teleconnections with Climate Indices: A Wavelet Analysis Approach," in *2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, Bhubaneswar, India, 2025. doi: 10.1109/InGARSS67683.2025.11584066.
2. **B. Naveen** and S. Upadhyaya, "Characterization of Precipitation Types over India Using GPM-DPR: Spatial Distribution, Vertical Structure, and Environmental Controls," *2026 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2026. **Accepted for presentation.**

Manuscript in Preparation

1. **B. Naveen** and S. Upadhyaya, "Physics-Aware Deep Learning for Hierarchical Multi-Class Precipitation Classification Using INSAT-3DR," manuscript in preparation for submission to *IEEE Transactions on Geoscience and Remote Sensing*.

ACADEMIC EXPERIENCE

Teaching Assistant – Data Modeling in Atmospheric Science IIT Hyderabad — 2025 – 2026

- Assisted with atmospheric-data analysis, climate datasets, environmental modelling, and Python-based scientific computing.
- Guided students in processing, visualization, and interpretation of atmospheric and environmental datasets.

CERTIFICATIONS

ESRI Spatial Data Science & Deep Learning | Machine Learning & Data Science Bootcamp – Udemy | Python Bootcamp – Udemy